§1-6

Def

1. E is finite $\Leftrightarrow \exists f: \{1,2,\cdots,n\} \to E$ is 1-1 and onto.
2. E is countable $\Leftrightarrow \exists f: \mathbb{N} \to E$ is 1-1 and onto.
3. E is at most countable $\Leftrightarrow E$ is finite or countable.
4. E is uncountable $\Leftrightarrow E$ is not at most countable

eg. 1. Countable: $\mathbb{Z}$, $\mathbb{Q}$

2. Uncountable: $(0,1)$, $\mathbb{R}$, $\mathbb{R}\setminus\mathbb{Q}$

① $|E| < |\mathbb{N}|$: E is finite.

② $|E| = |\mathbb{N}|$: E is countable.

③ $|E| > |\mathbb{N}|$: E is uncountable

1.40 ***Lemma.*** $\Leftrightarrow$ $\exists g: \mathbb{N} \to E$ is onto.
A nonempty set E is at most countable if and only if there is a function g from $\mathbf{N}$ onto E.

1.41 Theorem. *Suppose that A and B are sets.*

i) *If $A \subseteq B$ and B is at most countable, then A is at most countable.*
ii) *If $A \subseteq B$ and A is uncountable, then B is uncountable.*
iii) $\mathbf{R}$ *is uncountable.*

1.42 Theorem. *Let $A_1, A_2, \ldots$ be at most countable sets.*

i) *Then $A_1 \times A_2$ is at most countable.*
ii) *If*

$$E = \bigcup_{j=1}^{\infty} A_j := \bigcup_{j \in \mathbf{N}} A_j := \{x : x \in A_j \quad \text{for some } j \in \mathbf{N}\},$$

then E is at most countable.

1.6.0. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples to the false ones.

a) Suppose that E is a set. If there exists a function f from E onto $\mathbf{N}$, then E is at most countable.

b) A dyadic rational is a point $x \in \mathbf{R}$ such that $x = n/2^m$ for some $n \in \mathbf{Z}$ and $m \in \mathbf{N}$. The set of dyadic rationals is uncountable.

c) Suppose that A and B are sets and that $f : A \to B$ is 1–1. If A is uncountable, then B is uncountable.

d) If $E_1, E_2, \ldots$ are finite sets, and

$$E := E_1 \times E_2 \times \cdots := \{(x_1, x_2, \ldots) : x_j \in E_j \quad \text{for all } j \in \mathbf{N}\},$$

then E is countable.

(b) Let $D = \{\frac{n}{2^m} \mid n \in \mathbb{Z},\ m \in \mathbb{N}\}$. $\quad \mathbb{N} \to \mathbb{Z} \times \mathbb{N} \to D$

Define $f : \mathbb{Z} \times \mathbb{N} \to D \Rightarrow f(n,m) = \frac{n}{2^m}$

$\Rightarrow f(n,m)$ is onto

Since $\mathbb{Z}$ is at most countable $\Rightarrow \mathbb{Z} \times \mathbb{N}$ is at most countable

$\Rightarrow |D| \le |\mathbb{Z} \times \mathbb{N}|$

$\Rightarrow D$ is at most countable ~~✗~~

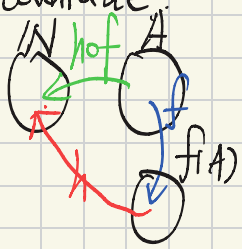
Hence it is false. $\quad f : A \to B \Rightarrow f(A) \subseteq B$

(c) Suppose B is countable, then $f(A)$ is countable.

Thus $f : A \to f(A)$ is bijective.

That is $\exists h : f(A) \to \mathbb{N}$ is onto

$\Rightarrow h \circ f : A \to \mathbb{N}$ is onto



By Lemma, A is at most countable. ~~✗~~

Hence B is uncountable.

It is true.

1.6.1. Prove that the set of odd integers $\{1, 3, \ldots\}$ is countable.

$f(n) = 2n-1, \quad n \in \mathbb{N}$

1.6.2. Prove that set of rational lattice points in space—that is, the set $\mathbf{Q}^3 := \{(x, y, z) : x, y, z \in \mathbf{Q}\}$—is countable.

$\mathbb{Q}^3 = \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}$

pf Since $\mathbb{Q}$ is countable,

by Thm 1.42(i), $\mathbb{Q} \times \mathbb{Q}$ is countable.

Therefore, $\mathbb{Q}^3 = (\mathbb{Q} \times \mathbb{Q}) \times \mathbb{Q}$ is countable.

Hence $\mathbb{Q}^3$ is countable. //

1.6.5. Let $f : A \to B$ and $g : B \to C$ and define $g \circ f : A \to C$ by $(g \circ f)(x) := g(f(x))$.

a) Show that if f, g are 1–1 (respectively, onto), then $g \circ f$ is 1–1 (respectively, onto).

b) Prove that if f is 1–1 from A into B and $B_0 := \{y : y = f(x) \text{ for some } x \in A\}$, then f^{-1} is 1–1 from B_0 onto A.

c) Suppose that g is 1–1 from B onto C. Prove that f is 1–1 on A (respectively, onto B) if and only if $g \circ f$ is 1–1 on A (respectively, onto C).

$gof: A \to C$

(a) ① 1-1 : Suppose $x_1, x_2 \in A$,

and $(gof)(x_1) = (gof)(x_2)$

Thus $(gof)(x_1) = (gof)(x_2) \Rightarrow g(f(x_1)) = g(f(x_2))$

$\Rightarrow f(x_1) = f(x_2)$

$\Rightarrow x_1 = x_2$

Hence gof is 1-1.

② onto :

$\forall z \in C$, since g is onto, $\exists y \in B$ s.t. $g(y) = z$.

$\forall y \in B$, since f is onto. $\exists x \in A$ s.t. $f(x) = y$

$\Rightarrow g(y) = g(f(x))$ is onto.

(c) ($\Rightarrow$) Since g is 1-1, by (a), gof is 1-1.

($\Leftarrow$) Since gof is 1-1 and g is 1-1,

Let $x_1, x_2 \in A$ s.t. $f(x_1) = f(x_2)$

Then $g(f(x_1)) = g(f(x_2)) \Rightarrow gof(x_1) = gof(x_2)$

$\Rightarrow x_1 = x_2 \Rightarrow f$ is 1-1. //

Onto:

($\Rightarrow$) By (a), $g \circ f$ is onto.

($\Leftarrow$) $\forall y \in B$. Consider $g(y) \in C$.

Since $g \circ f$ is onto, $\exists x \in A$ s.t. $\forall y \in B$

$(g \circ f)(x) = g(y) \Leftrightarrow g(f(x)) = g(y)$

Since g is 1-1, $f(x) = y$, hence f is onto. //

(b) ① Prove $B_0 = f(A)$

"$\subseteq$": $\forall y \in B_0$, $\exists x \in A$ s.t. $y = f(x) \in f(A)$, $B_0 \subseteq f(A)$

"$\supseteq$": $\forall y \in f(A)$, $\exists x \in A$ s.t. $y = f(x)$

By the def. of B_0, $y \in B_0 \Rightarrow B_0 \supseteq f(A)$

Hence $B_0 = f(A)$

② Prove $f^{-1}: B_0 \to A$ is 1-1 and onto

It is clear that $f: A \to f(A) = B_0$ is bijective, f^{-1} exists and $f^{-1}: B_0 \to A$ is bijective. //